

## Continuous polynomial graphs

A polynomial graph has no breaks or jumps.

**Could a graph with a hole represent a polynomial function on all real inputs?**

## Smooth polynomial graphs

A polynomial graph has no sharp corners.

**Could the graph of the absolute value function be a polynomial graph?**

## Finding zeros by factoring

$$f(x) = x^3 - 4x^2 - 5x$$

**What are all its real zeros?**

## Zeros from factored form

$$f(x) = 2(x + 3)(x - 1)(x - 4)$$

**Where does the graph meet the horizontal axis?**

## Multiplicity

A zero's multiplicity is the number of repetitions of its linear factor.

$$f(x) = (x + 2)^3(x - 5)^2$$

**What multiplicity does each zero have?**

## Odd multiplicity

$$f(x) = (x - 2)^3$$

**Does the graph change sign at  $x = 2$ ?**

## Even multiplicity

$$f(x) = -(x + 1)^4$$

**Does the graph cross or touch the horizontal axis at  $x = -1$ ?**

## Mixed multiplicities

$$f(x) = (x + 2)^2(x - 1)^3$$

**At which zero does the graph cross, and at which does it touch?**

## A vertical intercept

$$f(x) = -(x + 2)(x - 1)^2$$

**What is the vertical intercept?**

## End behavior from factors

$$f(x) = -2(x + 1)^2(x - 3)^3$$

**What happens at each end of the graph?**

## Intervals of positive values

$$f(x) = (x + 2)(x - 1)(x - 3)$$

**On what open intervals is  $f(x)$  positive?**

## Intervals of negative values

$$f(x) = (x + 1)^2(x - 2)$$

**On what open intervals is  $f(x)$  negative?**

## A repeated zero and a sign chart

$$f(x) = (x - 1)^2(x - 4)$$

**Why does  $x = 1$  split the sign chart without changing the sign?**

## Sketching a polynomial

$$f(x) = -(x + 2)(x - 1)^2$$

**What sketch matches its intercepts, multiplicities, and end behavior?**

## A higher-degree sketch

$$f(x) = (x + 1)^3(x - 2)^2$$

**What sketch matches the zeros and both ends?**

## Inferring a minimum degree

A polynomial graph crosses at  $-2$ , touches at  $1$ , and crosses at  $4$ .

**What is the smallest degree consistent with these behaviors?**

## Turning points

A turning point changes a graph from increasing to decreasing or the reverse.

**Must every horizontal-axis crossing be a turning point?**

## Counting possible turning points

$$f(x) = (x - 1)^2(x + 2)^2$$

**What is the maximum possible number of turning points?**

## Local maxima and minima

A local maximum or minimum compares a function value with values at nearby inputs.

**How does a local maximum differ from an absolute maximum?**

## Reading extrema with a utility

$$f(x) = x^3 - 3x$$

**Where does a graphing utility locate the local maximum and local minimum?**

## Increasing and decreasing intervals

Use the graphing utility's graph of:

$$f(x) = x^3 - 3x$$

**On what intervals is the function increasing or decreasing?**

## Intermediate Value Theorem

A polynomial takes every value between its outputs at the ends of an interval.

$$f(1) = -3, \quad f(2) = 5$$

**What can we conclude about a real zero between 1 and 2?**

## Locating a zero

$$f(x) = x^3 - x - 1$$

**What do  $f(1)$  and  $f(2)$  establish about a zero?**

## The theorem's limitation

$$f(x) = (x - 1)^2$$

**Do positive values at  $x = 0$  and  $x = 2$  rule out a zero between them?**